has picked up the master state, and that gratuitous ARPs have been propagated.

We are done with configuration for one cluster member, and have confirmed that all the services have been successfully started. You can now go ahead and configure the other peer in the same way, with only the sync IP different in */etc/conntrackd/conntrackd.conf*.

Before we go live...

After you finish with the configuration on the other peer, and before you proceed with the policy compilation, it is imperative to hold back a minute, and validate the changes. Let's do some testing and verify if the VRRP is working fine, and picking up the virtual IP addresses upon failure.

To verify that the *keepalived* and *conntrackd* daemons are running on both the peers, use the following code:

```
# service keepalived status
keepalived (pid 1374) is running...

# ps -aef | grep conntrackd
root      1406     1  0 23:01 ?        00:00:00 conntrackd -d
```

To identify which one is the master with the command *ip addr list*; you will see VRRP IP addresses attached to the master:

```
[root@Firewall2 ~]# ip addr list
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 16436 qdisc noqueue state UNKNOWN
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state
UNKNOWN qlen 1000
    link/ether 00:0c:29:d1:e8:d4 brd ff:ff:ff:ff:ff:ff
    inet 192.168.170.253/24 brd 192.168.170.255 scope global eth0
    inet 192.168.170.254/32 scope global eth0
    inet6 fe80::20c:29ff:fed1:e8d4/64 scope link
       valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state
UNKNOWN qlen 1000
    link/ether 00:0c:29:d1:e8:de brd ff:ff:ff:ff:ff:ff
    inet 172.16.1.253/24 brd 172.16.1.255 scope global eth1
    inet6 fe80::20c:29ff:fed1:e8de/64 scope link
       valid_lft forever preferred_lft forever
4: eth2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state
UNKNOWN qlen 1000
    link/ether 00:0c:29:d1:e8:e8 brd ff:ff:ff:ff:ff:ff
    inet 10.1.1.253/24 brd 10.1.1.255 scope global eth2
    inet 10.1.1.254/32 scope global eth2
    inet6 fe80::20c:29ff:fed1:e8e8/64 scope link
       valid_lft forever preferred_lft forever
```

Note: Firewall2 has picked up the master state, since we have specified a priority higher than that for Firewall1

Now, either bring down one of the interfaces on Firewall2, or stop the *keepalived* service on it, and see if Firewall1 is able to pick up those IP addresses:

```
[root@Firewall2 ~]# ifdown eth2

[root@Firewall1 ~]# tail -f /var/log/messages
May 22 22:54:12 localhost avahi-daemon[848]: Withdrawing address record for
192.168.170.254 on eth0.
May 22 23:13:30 localhost Keepalived_vrrp: VRRP_Instance(VI_2) Transition
to MASTER STATE
May 22 23:13:30 localhost Keepalived_vrrp: VRRP_Group(G1) Syncing instances
to MASTER state
May 22 23:13:30 localhost Keepalived_vrrp: VRRP_Instance(VI_1) Transition
to MASTER STATE
May 22 23:13:30 localhost Keepalived_vrrp: VRRP_Instance(VI_1) Entering
MASTER STATE
May 22 23:13:30 localhost Keepalived_vrrp: VRRP_Instance(VI_1) setting
protocol VIPs.
May 22 23:13:30 localhost avahi-daemon[848]: Registering new address record
for 192.168.170.254 on eth0.IPv4.
May 22 23:13:30 localhost Keepalived_vrrp: VRRP_Instance(VI_1) Sending
gratuitous ARPs on eth0 for 192.168.170.254
May 22 23:13:32 localhost Keepalived_vrrp: VRRP_Instance(VI_2) Entering
MASTER STATE
May 22 23:13:32 localhost Keepalived_vrrp: VRRP_Instance(VI_2) setting
protocol VIPs.
May 22 23:13:32 localhost avahi-daemon[848]: Registering new address record
for 10.1.1.254 on eth2.IPv4.
May 22 23:13:32 localhost Keepalived_vrrp: VRRP_Instance(VI_2) Sending
gratuitous ARPs on eth2 for 10.1.1.254
May 22 23:13:35 localhost Keepalived_vrrp: VRRP_Instance(VI_1) Sending
gratuitous ARPs on eth0 for 192.168.170.254
May 22 23:13:37 localhost Keepalived_vrrp: VRRP_Instance(VI_2) Sending
gratuitous ARPs on eth2 for 10.1.1.254
```

The *syslog* messages on Firewall1 clearly show that it has acquired master state. To really verify this, run *ip addr list* on Firewall1, and see if the VRRP IP addresses are attached to it.

This indicates that VRRP fail-over is running fine. Now, bring up the network interface on Firewall2, which will again cause a fail-over, and Firewall2 will again be switched over to master state.

Once we have verified that both the nodes are configured properly, let's proceed with the *fwbuilder* configuration.

Fwbuilder

Firewall Builder (also known as *fwbuilder*, http://www.fwbuilder.org) is (as per the website) a universal firewall configuration and